Mars MA3 SoC Module

Known Issues and Changes

Revision 2.0

Purpose

The purpose of this document is to describe the known issues and the revision changes related to the Mars MA3 SoC module.

Product Information	Code	Revision	Name
Product	MA-MA3	Mars MA3 R2.0	Mars MA3 SoC Module

Document Information	Reference	Version	Date	Author
Reference / Version / Date	D-0000-451-001	05	17.02.2021	DIUN

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1 Known Issues

In addition to the erratas from the components manufacturers, Enclustra is aware of the following issues with the Mars MA3 R2.0 module.

Booting from SD card on Mars PM3 baseboard is not fully supported	
Description	There are limitations when using the Mars MA3 SoC module in combination with the Mars PM3 base board. By default, on power-up, the Mars MA3 SoC module will try to boot from the QSPI flash (even with SD card boot settings), but after pressing the POR button the module will boot as expected from the device configured by the boot selection pins. QSPI flash boot works normally. This issue is related to how the bootstrap pins of the Cyclone V device are sampled in relation to the power sequence of the baseboard at start-up. If the SD card boot issue affects the target application, it is recommended to use a different Enclustra baseboard.
Workaround	Use a different Enclustra baseboard if possible.
Status	This issue will not be fixed.

Booting from SD card when VCC_CFG_HPS_B3A is 1.8 V is not fully supported	
Description	The SDIO level shifters, used for switching between the eMMC flash interface and the SD card interface available on the module connector, do not support a voltage of 1.8 V on VCCB pin (VCC_CFG_HPS_B3A). Enclustra tests have shown that the SoC device can boot at 1.8 V, but this voltage is out of the recommended range for the level shifter.
Workaround	Use the other accepted voltages (2.5 V or 3.3 V) on VCC_CFG_HPS_B3A.
Status	This issue will be fixed in the next revision (R3).

USB interface limitations	
Description	The USB interface on the Mars MA3 SoC module configured as host does not work in combination with a USB 3.0 capable device. In this case it was observed that USB power is disabled (CPEN is low). If a USB 2.0 capable device is used, the interface works as expected.
Workaround	Connect a USB 2.0 capable device to the Mars MA3 SoC module.
Status	This issue is still being investigated on Enclustra side. If a solution is found, this will be implemented in the next revision (R3).

Pull-ups missing on MDIO signals of the Fast Ethernet PHY	
Description	Part of the first production batch of Mars MA3 SoC modules is missing pull-up resistors on the MDIO lines of the Fast Ethernet interface, on the PHY side. The Fast Ethernet interface has been proven to function properly under laboratory conditions even without these pull-up resistors. The remaining R2 modules will be patched accordingly (revision R2.1, same EN number as R2.0) - refer to Figure 1 to find out if a specific module is patched. For better signal integrity it is also recommended activating the FPGA internal pull-ups on the Fast Ethernet MDIO signals (package pins AA18, AA19).
Workaround	If the interface has an unstable behavior, the MDIO signal lines may require the patch mentioned above. Please contact Enclustra Support for details.
Status	This issue affects only a part of the first production batch. Future R2 modules will be equipped with a patch that can be observed on the bottom right side of the module, near the mounting hole (Figure 1). This issue will be fixed on the PCB in the next revision (R3).

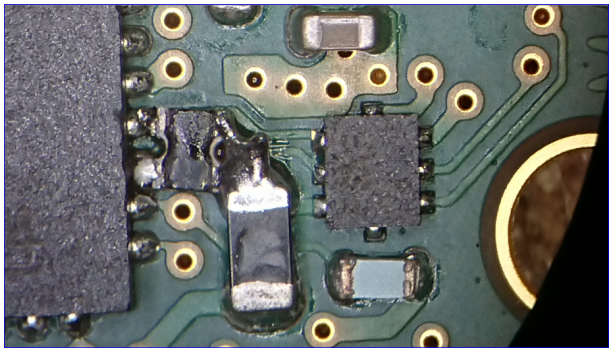

Figure 1: MDIO Pull-Up Resistors Patch (PHY side) - Bottom View (lower right part)

2 Functional Changes

The following functional changes were made from the Mars MA3 SoC module revision R1 to R2.0.

Changed FPGA clock frequency from 25 MHz to 50 MHz	
Description	The main clock for the FPGA logic (CLK_FPGA signal, package pin D12) is now 50 MHz, instead of 25 MHz.

Changed Fast Ethernet interface clocking structure	
Description	The 50 MHz oscillator clock is now driven directly to both FPGA logic (CLK_FPGA signal, package pin D12) and to the Fast Ethernet PHY (CLK_ETH1 signal), in order to obtain an RMII compliant clocking structure and ease timing closure for the FPGA. Please refer to the Mars MA3 SoC module user manual and schematics for details.

Optimized Fast Ethernet PHY bootstraps	
Description	The Fast Ethernet bootstraps have been improved to make sure the PHY settings are correctly set, regardless of the FPGA programming status. The PHY can be released from reset anytime before and after programming the FPGA, without affecting the configuration of the PHY (address and capabilities).

Separated MDIO bus for Gigabit and Fast Ethernet interfaces	
Description	The MDIO interface for the Fast Ethernet interface is now available from the FPGA side instead of HPS. The MDIO bus is completely separated for the Fast and Gigabit Ethernet PHYs. For Gigabit Ethernet, the MDIO remains connected to the HPS pins, like in revision 1 modules. The new path for the MDIO interface for the Fast Ethernet introduces level shifters, which only allow voltages equal to or higher than 1.8 V on the VCC_IO_B3B_B4A pin.

Added I2C bus connection to the PL	
Description	The I2C bus is now connected to both HPS and FPGA sides, for extra flexibility and future development. I2C on FPGA side is functional only when the VCC_IO_B3B_B4A voltage is 1.8 V or higher.